END-SEM (2026)

Biophysics(BIO215) **Total Marks:40**

Time: 2 hours

Section-A (MCQ) (1 mark each)

1. A scientist replaces several buried amino acids in the interior of a folded globular protein with glycine. After this mutation, the protein shows increased flexibility and reduced structural stability. What is the most appropriate explanation?
   a. Glycine promotes formation of additional disulfide bonds, disturbing the native protein structure
   b. Glycine introduces strong electrostatic repulsion within the hydrophobic core
   c. Glycine has a very small side chain, which increases backbone conformational flexibility and weakens stable packing.
   d. Glycine strengthens hydrophobic interactions inside the protein core, making folding less favorable

2. A 16-residue peptide sequence is predicted by a sliding-window secondary structure prediction method to form an α-helix because it contains several helix-favoring residues. However, experimental structures show that the same sequence forms an α-helix in one protein and a β-strand in another. Which conclusion is most appropriate?

   a. The prediction method is limited because it considers mainly local sequence preference and ignores the effect of the surrounding structural context.
   b. The prediction method is correct because helix-favoring residues can never form β-strands.
   c. The peptide must have undergone a mutation before forming a β-strand in the second protein.
   d. The β-strand structure must be impossible because the sequence was predicted as an α-helix.

3. In a molecular dynamics simulation, a researcher needs to calculate non-bonded interactions for millions of atom pairs at every time step. Although the Buckingham potential gives a more realistic description of short-range repulsion, the researcher chooses the Lennard–Jones potential instead. What is the most likely reason?

   a. Lennard–Jones potential is preferred because it completely removes the attractive van der Waals term
   b. Buckingham potential is avoided because it cannot describe repulsive interactions between atoms

c. Buckingham potential is avoided because it is only valid for covalent bonds, not non-bonded interactions
d. Lennard–Jones potential is preferred because its $1/r^{12}$ repulsive term is faster to compute than the exponential repulsive term in the Buckingham potential

4**.** After solvating a protein system in GROMACS, a .gro file is generated. What information is mainly stored in this file?

a. Force-field parameters and atom-type definitions
b. Atom coordinates along with simulation box dimensions
c. Energy minimization settings and cutoff values
d. Hydrogen-bond constraints and thermostat parameters

5. In most proteins, peptide bonds are usually found in the trans conformation, while the **cis** conformation is rarely observed. Which option best explains this observation and its common exception?

a. Cis peptide bonds are rare because they create steric clashes between adjacent Cα groups; proline is an exception where cis conformation occurs more frequently.
b. Cis peptide bonds are rare because they break the peptide bond completely; glycine is the only exception too
c. Trans peptide bonds are rare because they cause steric clashes between side chains; proline prevents trans peptide bond formation
d. Cis peptide bonds are common because peptide bonds rotate freely; proline completely blocks cis conformation

6. A researcher compares two homologous protein structures using Chimera Matchmaker. After alignment, Chimera reports a low RMSD value calculated over a large number of matched backbone atoms. What is the most appropriate interpretation?

a. The proteins have similar overall three-dimensional folds, although small local differences may still be present
b. The proteins must have identical amino acid sequences and identical biological functions
c. The proteins are structurally unrelated because low RMSD indicates poor superposition
d. The alignment is invalid because RMSD can only be calculated for DNA structures

7. After target-template alignment in MODELLER, a student receives both .ali and .pap output files. The student uses only the .pap file for model building and gets an error. What is the most likely reason?

a. .pap files contain atomic coordinates, so they cannot store alignments

b. .pap is mainly easier for visual inspection, while MODELLER uses PIR/ALI format for model building
c. .ali files are used only for Chimera visualization, not MODELLER
d. MODELLER accepts only FASTA files and never uses PIR/ALI alignment

8. A biophysicist is studying different motions in a protein and observes three processes:

1. Side-chain rotation occurring within a very small region
2. Hinge bending between two domains
3. Folding/unfolding of a large protein segment

Which option correctly matches these processes with their motion categories?

a. 1—Large-scale motion; 2—Local motion; 3—Rigid body motion
b. 1—Local motion; 2—Rigid body motion; 3—Large-scale motion
c. 1—Rigid body motion; 2—Large-scale motion; 3—Local motion
d. 1—Local motion; 2—Large-scale motion; 3—Rigid body motio

9. Match the GROMACS output file with its correct purpose:

1. em.gro
2. em.log
3. em.edr
4. em.trr

Which option is correct?

a. 1—Energy data, 2—Trajectory, 3—Final minimized structure, 4—Log file
b. 1—Final minimized structure, 2—Log file, 3—Energy data, 4—Trajectory file
c. 1—Topology file, 2—Energy curve, 3—Coordinate file, 4—Parameter file
d. 1—Input parameter file, 2—Final structure, 3—Topology metadata, 4—Index file

10. A researcher sets the Vina configuration file as:

center_x = 70, center_y = 70, center_z = 70
size_x = 25.5, size_y = 38.5, size_z = 29.6

Which statement best describes how these values affect the docking experiment?

a. To define the geometric center and spatial dimensions of the ligand search region in which Vina samples possible binding poses
b. To define the coordinates of every atom in the receptor and restrict ligand movement to covalent bond lengths only

c. To control the number of docking modes generated and rank them according to RMSD from the reference pose
d. To assign partial charges to receptor atoms and calculate electrostatic potential before docking

11. Doctor Strange traps a protein-ligand complex inside a time loop. Every loop begins with the same atomic positions, same velocities, same force field, and same temperature control. He notices that the atomic movie repeats exactly unless he changes the starting velocities. What principle is Strange exploiting?

a. Docking stochasticity, because repeated simulations should always generate different ligand poses even with the same starting coordinates.
b. MD determinism, because if initial positions, velocities, and forces are the same, Newtonian integration gives the same trajectory.
c. Boltzmann randomness, because lower-energy states are always reached instantly regardless of starting conditions.
d. Homology modelling, because similar protein sequences force the ligand to follow the same path.

12. Ultron mutates only the hinge loop of a defense enzyme so that the active site opens at the wrong time. The Avengers run MD and see that the overall protein RMSD is not huge, but residues 140–165 show very high fluctuation. Which analysis exposed Ultron's exact sabotage region?

a. H-bond analysis, because broken stabilizing interactions near the hinge may explain why the gate opens, but it does not directly quantify residue-wise flexibility.
b. RMSF, because it measures fluctuation residue by residue and can pinpoint residues 140–165 as unusually mobile.
c. Radius of gyration, because it reports whether the whole protein becomes more expanded, but it cannot localize one flexible hinge loop.
d. Potential energy analysis, because unstable systems may have high energy, but the energy curve mainly reports global relaxation/stability rather than a specific residue window.

13. Which thermodynamic principle primarily drives the burial of hydrophobic residues in a protein's core during folding?

a. Enthalpy minimization from van der Waals contacts
b. Gain in entropy of the surrounding water molecules
c. Electrostatic attraction between nonpolar side chains
d. Formation of covalent disulfide bonds in the core

14. Which of the following is the correct Open Babel command to convert all SDF files in the current directory into PDBQT format?

a. obabel -isdf *.sdf -opdb -O*.pdbqt
b. obabel -isdf *.sdf -opdbqt -O*.pdbqt
c. obabel -ipdb *.pdb -osdf -O*.sdf
d. obabel -isdf *.sdf -opdbqt -O*.pdb

15. What is the impact of using the -conc flag when adding an ion in genion?

a. It replaces protein residues with ions
b. It adds ions randomly
c. It ensures fixed neutralizing ions only
d. To add user-defined concentrations of ions beyond neutrality.

Section-B (2 mark each)

1. Humans possess two related proteins — Myoglobin (stores oxygen in muscle) and Hemoglobin (transports oxygen in blood). Both evolved from the same ancestral oxygen-binding protein. Dogs also possess Myoglobin, which stores oxygen in their muscles just like in humans.

(a) What is the evolutionary relationship between Human Myoglobin and Dog Myoglobin? Justify. *(1 Mark)*

(b) What is the evolutionary relationship between Human Myoglobin and Human Hemoglobin? Justify. *(1 Mark)*

Ans1 (a) Human Myoglobin and Dog Myoglobin are Orthologs. They exist in two different species (humans and dogs), perform the same function (oxygen storage in muscle), and diverged from the same ancestral myoglobin gene after a speciation event separated the two lineages.

(b) Human Myoglobin and Human Hemoglobin are Paralogs. Both exist within the same genome (humans), but arose from an ancestral oxygen-binding gene that underwent gene duplication. After duplication, the two copies diverged — one specialized for oxygen storage (myoglobin) and the other for oxygen transport (hemoglobin).

2. For the given config file below, identify the error and correct them to generate 15 mode of orientation while docking.

```
protein = hsg1.pdb
lig = ligand.sdf

center_x =  2
center_y =  6
center_z = -7

size_x = 25
size_y = 25
size_z = 25
```

```
receptor = hsg1.pdbqt
ligand = lisuride.pdbqt

center_x = 2
center_y = 6
center_z = -7

size_x = 25
size_y = 25
size_z = 25

num_modes = 15
```

3. The hydrophobic effect is considered the dominant driving force for protein folding into tertiary structure. Explain the thermodynamic basis of the hydrophobic effect does it arise from enthalpy, entropy, or both?

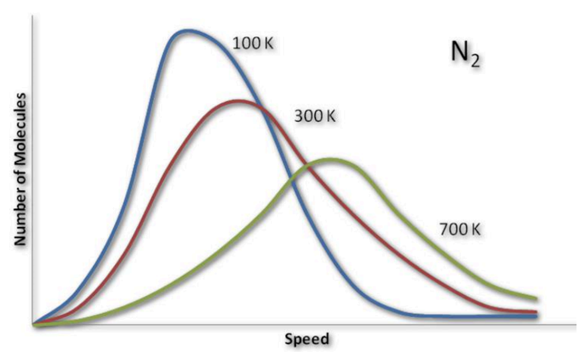
Ans: The hydrophobic effect is primarily entropy-driven. When nonpolar residues are exposed to water, surrounding water molecules form ordered cage-like structures (clathrate shells), reducing the entropy of the solvent ($\Delta S < 0$, unfavorable). Burying hydrophobic residues into the protein core releases these ordered water molecules back into bulk solvent, greatly increasing system entropy ($\Delta S > 0$, favorable) — making ΔG negative and folding thermodynamically spontaneous. The enthalpic contribution (van der Waals packing in the core) is secondary and stabilizing but not the primary driver.

4. A computational biochemist runs two separate MD simulations — one of a pure antiparallel β-sheet peptide and one of a parallel β-sheet peptide — both solvated in TIP3P water under identical NPT conditions at 300K and 1 bar. After 100 ns of production MD, the radius of gyration plot shows gradual expansion only in the parallel system. Both sheets use N-H···O=C hydrogen bonds — so geometrically explain at the atomic level why the H-bond geometry in antiparallel sheets is linearly optimal while parallel sheet H-bonds are inherently strained.

Ans. In an antiparallel β-sheet, strands run in opposite directions (N→C and C→N). This orientation allows N-H and C=O groups to align directly across from each other, producing hydrogen bonds that are nearly linear (~180°) — the geometrically ideal angle for maximum electrostatic attraction and orbital overlap between donor and acceptor.

In a parallel β-sheet, both strands run in the same direction (both N→C). This forces the H-bonds to form at a diagonal/oblique angle (~160° or less), introducing geometric strain. A bent hydrogen bond has a larger donor-acceptor distance, which directly weakens the electrostatic interaction.

5.. The N_2 speed distribution curve shows three plots at 100 K, 300 K, and 700 K. A student claims: "At 700 K there are fewer molecules at the peak speed, so pressure must be lower at 700 K than at 100 K." Identify the conceptual error in this statement and provide the correct reasoning linking temperature, molecular speed, and pressure.



Ans: The error is confusing the peak height (molecules at the most probable speed) with total kinetic energy. At 700 K the curve is flatter and wider, fewer molecules sit exactly at the peak, but far more move at high speeds than at 100 K. The total N is constant. Pressure $p = nkT$ is directly proportional to T, so higher T → higher pressure. More molecules at high speeds means more frequent and more energetic wall collisions and pressure goes up, not down.

Section-C (5 mark each)

Q1.You have a ligand–protein complex with no known binding data for other molecules at this target. You want to compare the binding affinities of 10 candidate ligands at the same site.

How would you select these 10 molecules, and what steps would you follow to compare their binding affinities with the known ligand? Given the binding pocket is flexible and ligand binding may induce conformational changes, explain why molecular modeling and MD simulations would be important in this case, and how they would improve prediction reliability.

- Download the Protein–Ligand Complex
- Prepare the Protein
    - Remove the existing ligand.
    - Remove water molecules
    - Add polar hydrogens and assign charges.
    - Save the protein in `.pdbqt` format
- Define the Binding Site
    - Use the coordinates of the original ligand to define the docking grid. (0.5 marks)
    - Draw a grid box that covers the binding pocket properly.
    - Create a configuration file
- Select 10 Potential Ligand Molecules on the basis of binding affinity (1.5)
- Ligand Preparation
    - Download the 3D structures (SDF/MOL2/PDB) of the selected 10 molecules.
    - Convert each ligand to `.pdbqt` format
- Molecular Docking
    - Dock all 10 ligands and the original ligand into the prepared protein. (0.5 marks)
    - Use the same grid box settings for all ligands to ensure fair comparison.
- Analyze Docking Results

There is no need for molecular modeling (1 marks)

Molecular modeling and MD simulations are essential when the binding pocket is flexible, as they capture protein and ligand movements that static docking misses. MD allows observation of induced fit, stabilizes docked poses, and reveals key interactions over time. This improves binding affinity predictions, especially when experimental data is unavailable, by providing a more realistic and dynamic view of ligand binding.(1.5 marks)

2. A novel pathogenic bacterial protein has been sequenced. BLAST analysis reveals 42% sequence identity with a structurally characterized human enzyme in the PDB. A researcher decides to build a homology model of this protein to facilitate drug design. Critically analyze and answer the following:

a. At 42% sequence identity, the researcher is cautiously optimistic. Justify whether this level of identity is sufficient for reliable homology modeling and explain what "twilight zone" in sequence–structure relationships means in this context. (1 marks)

   Ans: 42% sequence identity is sufficient for homology modeling sequences above ~30% identity usually share similar 3D folds. The twilight zone is the range of 20–35% identity where it becomes uncertain whether two proteins share the same structure or not. At 42%, the template is reliable, though loop regions may still differ.

   [Award mark for: correct zone assessment + twilight zone explanation with ~20–35% range]

b. Outline the key steps in building a homology model using this template. Why is loop modeling considered the most challenging step, and what happens to model quality in regions of low sequence conservation? (1.5 marks)

   Modeling steps.

   Loop modeling is challenging because loops lack the hydrogen-bonding constraints of helices/sheets, exist in vast conformational space, and are most evolutionarily variable.

   In regions of low sequence conservation, the model has higher RMSD from the true structure, side-chain positions are unreliable

   [Award 1.5 for: correct ordered steps (0.5) + loop challenge explained (0.5) + low conservation impact (0.5)]

c. The researcher validates the model using a Ramachandran plot and DOPE score. The plot shows 86% residues in the favored region, and several loop residues fall in the disallowed region. Interpret these results, is this model acceptable for drug docking, and what steps would you recommend next? (1.5 marks)

   ans:86% residues in favored region is borderline — good models have >90%. Residues in the disallowed region indicate bad backbone geometry (steric clashes). If these fall in the binding site, the model is not suitable for docking directly.

   Recommended steps: Refine the loops, perform energy minimization, re-validate with ProSA or ERRAT, and confirm the active site passes before docking.

   [Award 1.5 for: correct interpretation of 86% (0.5) + disallowed region significance (0.5) + actionable next steps (0.5)]

d. Homology modeling assumes that evolutionarily related proteins with similar sequences adopt similar folds. How might this assumption fail when the template is a human enzyme (eukaryote) and the target is a bacterial protein (prokaryote)? Name one alternative computational approach if this assumption breaks down entirely.

   ans: The assumption may fail because the human (eukaryotic) template may have glycosylation, disulfide bonds, or need chaperones that the bacterial protein does not leading to different folding. If the assumption breaks down, AlphaFold2 (ab initio/deep learning-based prediction) can be used as it does not need a homologous template.

3. Ant-Man enters a protein-ligand MD simulation. The computer does not follow atoms continuously; it advances them in tiny femtosecond steps. At each step, forces are calculated from the force field, positions/velocities are updated, and the process repeats.

When Ant-Man changes the timestep from **2 fs to 2 ps**, bonds stretch unrealistically, atoms overlap, and the protein "explodes." Wasp says: "You broke the integrator, not the protein."

Answer the following (1 mark each):

A. How does MD convert force into atomic motion?
B. What does the Verlet update use to predict the next position?
C. What is the basic idea of Leapfrog Verlet?
D. Why does a large timestep make the simulation unstable?
E. What is a trajectory and why is it useful?

Ans:

A. Force to motion: MD uses Newton's second law, F=ma. The force field calculates forces from potential energy, and acceleration is calculated as a=F/m

B. Verlet update:
Verlet predicts the next position using current position, previous position, and current acceleration:

r(t+Δt)=2r(t)−r(t−Δt)+a(t)(Δt)2r(t+\Delta t)=2r(t)-r(t-\Delta t)+a(t)(\Delta t)^2r(t+Δt)=2r(t)−r(t−Δt)+a(t)(Δt)2

C. Leapfrog Verlet:
In Leapfrog Verlet, velocity is calculated at half-time steps and position at full-time steps, so velocity and position "leapfrog" over each other.

D. Timestep issue:
Atoms vibrate on femtosecond timescales. If Δt\Delta tΔt is too large, fast force changes are missed, causing unrealistic jumps, atom overlaps, and simulation instability.

E. Trajectory:
A trajectory is the time-series record of atomic positions/velocities. It is used to analyze RMSD, RMSF, radius of gyration, hydrogen bonds, and stability.